

WASP-10b

R-2145

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_111123 · created 2026-08-12T00:38:02+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455888.75643 +/- 0.00071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.011
Transit depth (Rp/Rs)^2	2.82 +/- 0.36 [%]
Orbital Inclination (inc)	86.92 +/- 0.76
Airmass coefficient 1 (a1)	1.0083 +/- 0.0099
Airmass coefficient 2 (a2)	-0.0069 +/- 0.003
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#3 - [168, 253]
Optimal Method	PSF photometry
Transit Duration (day)	0.0806 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.168 ± 0.011 vs 0.1592 (deviation 0.7σ)
Duration fit vs published	0.0806 d vs 0.0946 d (deviation 1.5σ)
Mid-time O-C	-3.1 min
Depth SNR	13.3
Residual scatter	0.99 %

Light curve

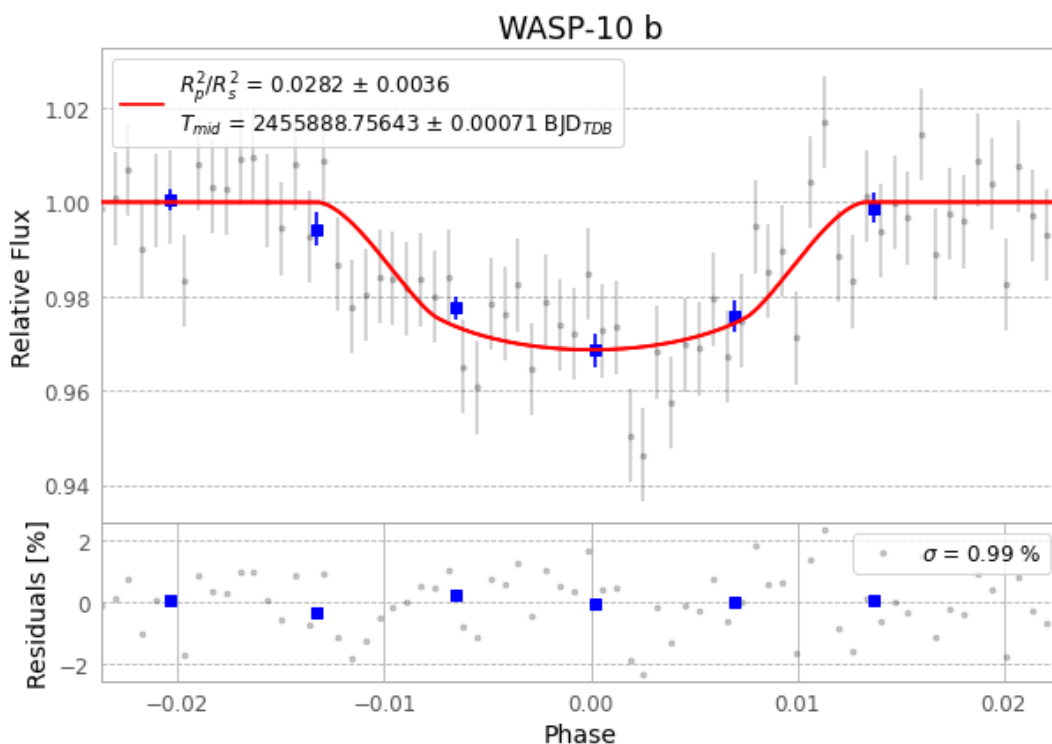


Figure 1 — FinalLightCurve_WASP-10 b_2011-11-22.png (SHA-256 036ea7db229a8a54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f7f4fa57a7843de6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), WASP-10111123041815.FITS → WASP-10111123074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-111123.log (a2b0a88eff93...), FinalLightCurve_WASP-10 b_2011-11-22.png (036ea7db229a...), FinalLightCurve_WASP-10 b_2011-11-22.csv (cf10f6ecf905...)

Compute resources

Wall time 141.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:38Z.